Olympiad Test Paper — Mathematics (Class 7) — Paper 3

Chapter 5: Parallel and Intersecting Lines

Paper 3 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	5 — Parallel and Intersecting Lines
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Treating corresponding/alternate angles as equal when the lines are NOT parallel; treating co-interior (same-side interior) angles as equal when they are actually supplementary; confusing alternate (equal) with co-interior (sum 180°); skipping a linear-pair step before applying a parallel-line relation; stopping the angle chase one step early; using 360° where 180° applies (and the reverse). Do not write on this paper — mark answers on your answer sheet.

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1. Two lines AB and CD cross at O. Angle AOC = $(4x)^\circ$ and the adjacent angle BOC = $(5x)^\circ$ form a linear pair. Find angle AOC.

(A) 100° (B) 72°
(C) 20° (D) 80°

2. Two lines intersect at O. Angle AOC = $(3x + 12)^\circ$ and its vertically opposite angle BOD = $(5x - 18)^\circ$. Find angle AOC.

(A) 57° (B) 123°
(C) 15° (D) 78°

3. At an intersection of two lines, one angle is $(4x - 6)^\circ$ and its adjacent angle (linear pair) is $(2x + 6)^\circ$. Find the larger of these two angles.

(A) 66° (B) 30°
(C) 114° (D) 120°

4. A transversal cuts two parallel lines p and q. A corresponding angle at p is $(2x + 30)^\circ$ and the matching corresponding angle at q is $(4x - 10)^\circ$. Find that common corresponding-angle measure.

(A) 70°
(B) 20°
(C) 110°
(D) 50°

5. Line a is perpendicular to line b, and line c is parallel to line b. A transversal makes a 35° angle with line a. What acute angle does that transversal make with line c?

(A) 35°
(B) 90°
(C) 125°
(D) 55°

6. Lines l and m are parallel, cut by a transversal. A pair of alternate interior angles measure $(3x - 5)°$ and $(x + 41)°$. Find the measure of each of these alternate angles.

(A) 23°
(B) 64°
(C) 116°
(D) 46°

7. Three rays from a point O divide the full turn so that the angles are $(2x)°$, $(3x + 10)°$, and $(4x - 10)°$ going once around. Find the largest of these three angles.

(A) 80°
(B) 150°
(C) 130°
(D) 120°

8. A transversal cuts two parallel lines. One co-interior (same-side interior) angle is $(3x + 20)°$ and the other is $(2x + 10)°$. Find the larger co-interior angle.

(A) 70°
(B) 90°
(C) 110°
(D) 100°

9. Two lines intersect at O. One angle is $(x + 15)°$ and its vertically opposite angle is $(2x - 35)°$. Find the measure of the linear-pair neighbour of these angles.

(A) 65°
(B) 115°
(C) 50°
(D) 130°

10. A transversal cuts two parallel lines so that the eight angles take just two distinct values, and one value exceeds the other by 50°. Find the larger of the two values.

(A) 65°
(B) 90°
(C) 130°
(D) 115°

11. Lines u and v are cut by a transversal. A pair of alternate interior angles measure 70° (at u) and 76° (at v). What can you conclude about u and v?

(A) u and v are parallel
(B) u and v are not parallel
(C) u and v are perpendicular
(D) u and v are coincident

12. Lines l and m are parallel, cut by a transversal. A pair of alternate exterior angles measure $(5x - 30)°$ and $(3x + 10)°$. Find each alternate exterior angle.

(A) 20°
(B) 110°
(C) 70°
(D) 40°

13. A transversal cuts two parallel lines so that one co-interior angle is three times its same-side partner. Find the larger co-interior angle.

(A) 45°　　(B) 90°

(C) 120°　　(D) 135°

14. At an intersection of two lines, one angle is (5x – 9)° and its adjacent angle (linear pair) is (x + 9)°. Find the smaller of these two angles.

(A) 141°　　(B) 30°

(C) 51°　　(D) 39°

15. Lines p and q are parallel, cut by transversal t. Corresponding angles measure (6x – 8)° at p and (4x + 22)° at q. Find that corresponding-angle measure.

(A) 15°　　(B) 82°

(C) 98°　　(D) 67°

16. A transversal cuts two parallel lines u and v. Co-interior angles measure (3x)° at u and (2x)° at v. After finding x, the angle corresponding to the (3x)° angle (located at v) measures:

(A) 72°　　(B) 36°

(C) 108°　　(D) 144°

17. Which condition does NOT by itself guarantee that two lines cut by a transversal are parallel?

(A) A pair of alternate interior angles are equal

(B) A pair of corresponding angles are equal(C) A pair of co-interior angles sum to 180°

(D) A pair of co-interior (same-side interior) angles are equal

18. A transversal cuts two parallel lines so that one of the eight angles exceeds its linear-pair neighbour by 36°. Find the smaller of the two distinct values among the eight angles.

(A) 108°　　(B) 72°

(C) 90°　　(D) 54°

19. Two lines intersect so that one of the four angles equals the sum of the other three minus 180°. Using that the four angles total 360°, find that one angle.

(A) 180°　　(B) 120°

(C) 60°　　(D) 90°

20. A transversal cuts lines a and b. Corresponding angles are (4x + 10)° at a and (6x – 30)° at b. The lines are parallel only when these are equal. At x = 25 (not the parallel value), by how much do the two corresponding angles differ?

(A) 0°　　(B) 20°

(C) 5°　　(D) 10°

21. A transversal crosses line a at 90° (perpendicular). The same transversal crosses line b, which is parallel to a. One angle at b measures (5x – 5)°. Find x.

(A) 18 (B) 19
(C) 17 (D) 95

22. Lines l and m are parallel, cut by a transversal. An alternate interior angle is $(7x - 4)°$ and the alternate interior angle on the other line is $(5x + 14)°$. After finding the angle, its co-interior partner (same-side interior, at the same line) measures:

(A) 59° (B) 9°
(C) 121° (D) 63°

23. In a figure, AB and CD intersect at O. Ray OE lies inside angle BOD. Given angle AOC = 64° and angle BOE = 26°, find angle DOE.

(A) 90° (B) 38°
(C) 116° (D) 64°

24. A transversal cuts two parallel lines. A same-side interior (co-interior) angle is 18° more than its same-side partner. Find the smaller co-interior angle.

(A) 99° (B) 81°
(C) 90° (D) 72°

25. Two perpendicular lines meet at O. A ray OP from O makes 28° with one of the lines, lying inside one of the right angles. Find the angle OP makes with the OTHER of the two perpendicular lines in that same right angle.

(A) 28° (B) 90°
(C) 62° (D) 118°

26. Three angles meeting at a point O, going once around, are $(2y)°$, $(2y + 30)°$, and $(5y - 30)°$. Find the largest of the three angles.

(A) 170° (B) 80°
(C) 110° (D) 150°

27. Lines l and m are parallel, cut by transversal t. Angle 1 (interior, at l) corresponds to angle 2 (at m), and angle 3 is the linear pair of angle 2. If angle 1 = 133°, find angle 3.

(A) 133° (B) 90°
(C) 27° (D) 47°

28. A transversal cuts two parallel lines, and one of the eight angles is a right angle. How many of the eight angles are right angles?

(A) 8 (B) 1
(C) 2 (D) 4

29. Lines a and b are parallel, cut by transversal t. Corresponding angles are angle P = $(8x - 12)°$ at a and angle Q = $(5x + 21)°$ at b. After finding the angle, the linear-pair neighbour of angle Q measures:

(A) 76° (B) 11°
(C) 98° (D) 104°

30. Two lines intersect at O. The four angles in order are p, q, p, q with $p - q = 48°$. Find q.

(A) 114° (B) 66°

(C) 24° (D) 90°

31. Lines l and m are parallel, cut by a transversal. An alternate-interior angle and a co-interior angle share the same vertex on one line and form a linear pair there. The alternate-interior angle is 40° more than the co-interior one. Find the alternate-interior angle.

(A) 70° (B) 90°

(C) 110° (D) 140°

32. Two roads cross. The north-east corner angle is $(3x + 18)°$ and the adjacent north-west corner angle is $(2x + 12)°$; together they form a linear pair. Find the north-east angle.

(A) 72° (B) 30°

(C) 108° (D) 84°

33. A transversal cuts two lines. A pair of corresponding angles measure 102° (at the first line) and 102° (at the second line). What is the correct conclusion?

(A) The two lines are parallel (B) The two lines are perpendicular

(C) Nothing can be concluded (D) The two lines meet at 102°

34. Lines l and m are parallel, cut by transversal t. Angle 1 and angle 2 are co-interior angles, and their bisectors meet at a point. Given angle 1 = 100°, find the angle between the two bisectors.

(A) 90° (B) 50°

(C) 40° (D) 100°

35. A transversal cuts two parallel lines so that one alternate-interior angle and its co-interior partner at the SAME vertex are in ratio 5:4 (they form a linear pair). Find the alternate-interior angle.

(A) 80° (B) 90°

(C) 100° (D) 108°

36. A transversal cuts two lines. Co-interior angles measure 95° and 86°. Decide whether the lines are parallel and by how much the supplementary test fails.

(A) Parallel; they sum to 180°

(B) Not parallel; the angles sum to 181°, which is 1° over 180°

(C) Perpendicular; one angle is near 90° (D) Not parallel; they differ by 9°

37. In a figure, line PQ is parallel to line RS. A transversal meets PQ at A and RS at B. Angle QAB = $(2x + 25)°$ and angle ABR = $(3x + 5)°$ are alternate interior angles. Find angle QAB.

(A) 20°
(B) 115°
(C) 65°
(D) 85°

38. Lines m and n are parallel, cut by a transversal. Corresponding angles measure (x + 40)° at m and (3x – 20)° at n. After finding the angle, its alternate-interior angle on the other line measures:

(A) 70°
(B) 110°
(C) 30°
(D) 50°

39. A transversal cuts two parallel lines. An exterior angle on the upper line is 142°. Find the alternate interior angle on the lower line.

(A) 142°
(B) 38°
(C) 52°
(D) 48°

40. Two intersecting lines make four angles; reading around, three of them are (x)°, (2x – 30)°, and (x)°. Find x.

(A) 70
(B) 50
(C) 60
(D) 75

41. Lines AB and CD intersect at O. Angle AOC = 52°. Ray OE is drawn with angle AOE = 90° (OE on the opposite side of AB from C, same side as D). Find angle EOD.

(A) 38°
(B) 52°
(C) 128°
(D) 90°

42. A transversal cuts two parallel lines so that an alternate interior angle equals twice its co-interior partner at the same vertex (they form a linear pair). Find the alternate interior angle.

(A) 120°
(B) 60°
(C) 90°
(D) 135°

43. Which ordered set of four angle measures could be the four angles formed by two intersecting non-perpendicular lines, read around the point?

(A) 70°, 110°, 110°, 70°
(B) 90°, 90°, 90°, 90°
(C) 70°, 110°, 70°, 110°
(D) 70°, 70°, 110°, 110°

44. A transversal cuts two parallel lines. The eight angles take two values, one being twice the other. Find the smaller value.

(A) 120°
(B) 90°
(C) 60°
(D) 45°

45. A transversal cuts two parallel lines. The difference between the two co-interior angles is 24°. Find the larger co-interior angle.

(A) 78°
(B) 90°
(C) 114°
(D) 102°

46. A transversal cuts two parallel lines l and m. Angle A on l and angle B on m are alternate angles with angle A = (2·angle B – 24)°. Find angle A.

(A) 48° (B) 24°

(C) 12° (D) 156°

47. A transversal crosses two parallel lines at exactly 90°. How many distinct angle measures appear among the eight angles, and what are they?

(A) Exactly 1 distinct value: 90° (B) 2 distinct values: 90° and 90°

(C) 4 distinct values (D) 2 distinct values: 80° and 100°

48. Two intersecting lines make four angles; three of them, read around, are (3x)°, (x + 20)°, and (3x)°. Find x.

(A) 40 (B) 20

(C) 35 (D) 45

49. A transversal cuts two parallel lines so that one alternate-interior angle and its co-interior partner at the same vertex are in ratio 7:2 (they form a linear pair). Find the smaller (co-interior) angle.

(A) 140° (B) 40°

(C) 90° (D) 20°

50. Lines l and m are parallel, cut by transversal t. Angle 1 corresponds to angle 2; angle 2 forms a linear pair with angle 3; angle 3 is vertically opposite angle 4. If angle 1 = 124°, find angle 4.

(A) 124° (B) 90°

(C) 34° (D) 56°

51. A transversal cuts two lines. A pair of corresponding angles measure 88° at the first line and 92° at the second line. What can you conclude?

(A) The two lines are not parallel (B) The two lines are parallel

(C) The two lines are perpendicular (D) The two lines are coincident

52. Lines m and n are parallel, cut by a transversal. Corresponding angles measure (x + 25)° at m and (2x – 15)° at n. After finding the angle, its co-interior partner at the same crossing measures:

(A) 115° (B) 65°

(C) 40° (D) 130°

53. A transversal cuts two parallel lines so that an alternate-interior angle and a co-interior angle at the same vertex are in ratio 3:2 (they form a linear pair). Find the alternate-interior angle.

(A) 108° (B) 72°

(C) 90° (D) 120°

54. Two lines intersect at O. Angle AOC = $(4x + 5)°$ and its vertically opposite angle BOD = $(6x - 25)°$. After finding the angle, the linear-pair neighbour angle AOD measures:

(A) 65°　　(B) 115°

(C) 15°　　(D) 130°

55. Lines a and b are parallel, cut by a transversal. The bisector of an 84° corresponding angle is drawn. Find the acute angle this bisector makes with line b (the arm of the corresponding angle at b).

(A) 84°　　(B) 48°

(C) 42°　　(D) 138°

56. A transversal cuts two parallel lines. An alternate interior angle is $(2x + 9)°$ and the alternate interior angle on the other line is $(3x - 6)°$. Find each of these alternate angles.

(A) 15°　　(B) 141°

(C) 45°　　(D) 39°

57. Parallel lines l and m are cut by a transversal. Angle P is an interior angle at l equal to 73°, and angle Q is its co-interior partner at m. The bisector of angle Q is drawn. Find the acute angle this bisector makes with line m.

(A) 36.5°　　(B) 73°

(C) 107°　　(D) 53.5°

58. Two lines intersect at O. Angle AOC and angle COB form a linear pair with angle AOC = $(7x)°$ and angle COB = $(2x)°$. Ray OD bisects angle AOC. Find angle DOC.

(A) 140°　　(B) 40°

(C) 70°　　(D) 20°

59. A transversal cuts two parallel lines so that one co-interior angle is four times its same-side partner. Find the larger co-interior angle.

(A) 36°　　(B) 90°

(C) 144°　　(D) 135°

60. Lines l and m are parallel, cut by transversal t at P (on l) and Q (on m). At P an angle of 118° is formed above l. Find the angle at Q that is vertically opposite to the corresponding angle of that 118° angle.

(A) 62°　　(B) 90°

(C) 236°　　(D) 118°

Solutions — Mathematics (Class 7), Chapter 5: Parallel and Intersecting Lines — Paper 3

Paper 3 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	B	21	B	31	C	41	A	51	A
2	A	12	C	22	C	32	C	42	A	52	A
3	C	13	D	23	B	33	A	43	C	53	A
4	A	14	D	24	B	34	A	44	C	54	B
5	D	15	B	25	C	35	C	45	D	55	C
6	B	16	C	26	A	36	B	46	B	56	D
7	B	17	D	27	D	37	C	47	A	57	D
8	C	18	B	28	A	38	A	48	A	58	C
9	B	19	D	29	D	39	B	49	B	59	C
10	D	20	D	30	B	40	A	50	D	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) Angle AOC and angle BOC are a linear pair on line AB, so $4x + 5x = 180$, giving $9x = 180$, $x = 20$. Then angle AOC = $4x = 80°$. The 100° is the neighbour BOC, not AOC. The 72° wrongly sets $5x = 180$. The 20° is x itself, not the angle.

2. (A) Vertically opposite angles are equal, so $3x + 12 = 5x - 18$, giving $30 = 2x$, $x = 15$. Hence angle AOC = $3(15) + 12 = 57°$. The 123° is the supplementary linear-pair neighbour, not AOC. The 15° is x. The 78° comes from a mis-solve.

3. (C) A linear pair sums to 180°: $(4x - 6) + (2x + 6) = 180$, so $6x = 180$, $x = 30$. The angles are $4(30) - 6 = 114°$ and $2(30) + 6 = 66°$; the larger is 114°. The 66° is the smaller. The 30° is x. The 120° forgets the −6.

4. (A) Between parallel lines corresponding angles are equal: $2x + 30 = 4x - 10$, so $40 = 2x$, $x = 20$. The angle = $2(20) + 30 = 70°$. The 20° is x. The 110° is the supplement (a co-interior partner), the trap of treating these as supplementary. The 50° uses a wrong x in $4x - 10$.

5. (D) Since c is parallel to b and a is perpendicular to b, line a is also perpendicular to line c. The transversal makes 35° with a, so with c (which is 90° from a) it makes 90 − 35 = 55°. The 35° forgets the 90° turn. The 90° ignores the transversal. The 125° is the obtuse supplement, not the acute angle.

6. (B) Alternate interior angles between parallel lines are equal: 3x − 5 = x + 41, so 2x = 46, x = 23. Each angle = 3(23) − 5 = 64°. The 23° is x. The 116° is the supplement, wrong because alternate angles are equal, not supplementary. The 46° is 2x.

7. (B) Angles at a point sum to 360°: 2x + (3x + 10) + (4x − 10) = 360, so 9x = 360, x = 40. The angles are 80°, 130°, and 150°; the largest is 4(40) − 10 = 150°. The 80° is the smallest. The 130° is the middle one. The 120° wrongly uses a 180° total.

8. (C) Co-interior angles between parallel lines are supplementary: (3x + 20) + (2x + 10) = 180, so 5x + 30 = 180, 5x = 150, x = 30. The angles are 3(30) + 20 = 110° and 2(30) + 10 = 70°; the larger is 110°. The 70° is the smaller. The 90° wrongly assumes equality. The 100° is a setup error.

9. (B) Vertically opposite: x + 15 = 2x − 35, so x = 50, and the angle = 50 + 15 = 65°. Its linear-pair neighbour = 180 − 65 = 115°. The 65° is the angle itself, not its neighbour. The 50° is x. The 130° doubles 65 instead of subtracting from 180.

10. (D) The two values form linear pairs, so they sum to 180°. Let them be a and a + 50: a + (a + 50) = 180, 2a = 130, a = 65, larger = 115°. The 65° is the smaller. The 90° needs a 0° difference. The 130° is 2a, not the angle.

11. (B) If u and v were parallel, alternate interior angles would be equal. Here 70° ≠ 76°, so the lines cannot be parallel. 'Parallel' applies the equality without checking it holds. 'Perpendicular' needs a 90° angle, absent here. 'Coincident' would make the angles meaningless, not the case.

12. (C) Alternate exterior angles between parallel lines are equal: 5x − 30 = 3x + 10, so 2x = 40, x = 20. Each angle = 5(20) − 30 = 70°. The 20° is x. The 110° is the supplement, wrong since alternate angles are equal. The 40° is 2x.

13. (D) Co-interior angles are supplementary: let the smaller be a and the larger 3a. Then a + 3a = 180, 4a = 180, a = 45, larger = 3(45) = 135°. The 45° is the smaller. The 90° wrongly assumes equality. The 120° would come from a 2:1 ratio, not 3:1.

14. (D) Linear pair sums to 180°: (5x − 9) + (x + 9) = 180, so 6x = 180, x = 30. The angles are 5(30) − 9 = 141° and (30) + 9 = 39°; the smaller is 39°. The 141° is the larger. The 30° is x. The 51° is a slip.

15. (B) Corresponding angles are equal between parallel lines: 6x − 8 = 4x + 22, so 2x = 30, x = 15. The angle = 6(15) − 8 = 82°. The 15° is x. The 98° is the supplement (a co-interior trap). The 67° is a slip in solving.

16. (C) Co-interior angles are supplementary: $3x + 2x = 180$, $5x = 180$, $x = 36$. The $(3x)°$ angle is 108°. Its corresponding angle at v equals it (parallel lines): 108°. The 72° is the $(2x)°$ co-interior partner. The 36° is x. The 144° is 4x, unjustified.

17. (D) Parallelism requires co-interior angles to SUM to 180°, not to be equal; equal co-interior angles do not in general force the sum to be 180° (only the 90°-90° special case does). Equal alternate interior angles, equal corresponding angles, and co-interior angles summing to 180° each correctly guarantee parallelism. So the equal-co-interior condition is the one that fails as a general guarantee.

18. (B) An angle and its linear-pair neighbour sum to 180°. Let smaller = a, larger = a + 36: $a + (a + 36) = 180$, $2a = 144$, $a = 72°$. The 108° is the larger. The 90° needs a 0° gap. The 54° halves the difference wrongly.

19. (D) Let the angle be a; the other three sum to 360 − a. The condition gives $a = (360 - a) - 180 = 180 - a$, so $2a = 180$, $a = 90°$. The 180° forgets to halve. The 120° and 60° do not satisfy $a = 180 - a$.

20. (D) At $x = 25$: angle at a = $4(25) + 10 = 110°$, angle at b = $6(25) - 30 = 120°$. They differ by $120 - 110 = 10°$, so the corresponding angles are unequal and the lines are not parallel. The 0° would require $x = 20$ (the parallel value). The 20° doubles the gap. The 5° halves it.

21. (B) Since the transversal is perpendicular to a and b is parallel to a, the transversal is perpendicular to b too, so every angle at b is 90°. Thus $5x - 5 = 90$, $5x = 95$, $x = 19$. The 18 comes from $5x = 90$. The 17 from $5x - 5 = 80$. The 95 is 5x, not x.

22. (C) Alternate interior angles are equal: $7x - 4 = 5x + 14$, so $2x = 18$, $x = 9$. The angle = $7(9) - 4 = 59°$. Its co-interior (same-side) partner is supplementary: $180 - 59 = 121°$. The 59° is the alternate angle, not its co-interior partner. The 9° is x. The 63° is 7x, a slip.

23. (B) Angle BOD is vertically opposite angle AOC, so angle BOD = 64°. Ray OE lies inside it with angle BOE = 26°, so angle DOE = angle BOD − angle BOE = 64 − 26 = 38°. The 90° wrongly assumes OE is perpendicular. The 116° is 180 − 64, the linear pair of BOD. The 64° forgets to subtract OE's 26°.

24. (B) Co-interior angles are supplementary. Let the smaller be a, the larger a + 18: $a + (a + 18) = 180$, $2a = 162$, $a = 81°$. The 99° is the larger angle. The 90° needs a 0° difference. The 72° is a wrong subtraction.

25. (C) The two perpendicular lines form a 90° angle. OP lies inside it making 28° with one arm, so it makes $90 - 28 = 62°$ with the other arm. The 28° repeats the given. The 90° ignores OP. The 118° is 90 + 28, a wrong sum.

26. (A) Angles at a point sum to 360°: $2y + (2y + 30) + (5y - 30) = 360$, so $9y = 360$, $y = 40$. The angles are $2(40) = 80°$, $2(40) + 30 = 110°$, and $5(40) - 30 = 170°$; the largest is 170°. The 80° and 110° are the two smaller angles. The 150° comes from a slip in evaluating $5y - 30$.

27. (D) Angle 2 = angle 1 = 133° (corresponding angles, parallel lines). Angle 3 is the linear pair of angle 2: $180 - 133 = 47°$. The 133° forgets the linear-pair step. The 90° is unjustified. The 27° comes from $160 - 133$, an arithmetic slip.

28. (A) If the transversal makes a right angle with one parallel line it is perpendicular to it; since the lines are parallel, it is perpendicular to the other line too. Every one of the eight angles is then 90°, so all 8 are right angles. The 1, 2, and 4 each undercount; only NON-parallel crossings could give fewer right angles.

29. (D) Corresponding angles are equal: $8x - 12 = 5x + 21$, so $3x = 33$, $x = 11$. Angle Q = $5(11) + 21 = 76°$. Its linear-pair neighbour = $180 - 76 = 104°$. The 76° forgets the linear-pair step. The 11° is x. The 98° is a wrong subtraction.

30. (B) Adjacent angles p and q form a linear pair: $p + q = 180$. With $p - q = 48$, adding gives $2p = 228$, $p = 114$, so $q = 66°$. The 114° is p, not q. The 24° halves the difference incorrectly. The 90° ignores the difference.

31. (C) Sharing a vertex as a linear pair, the two sum to 180°. Let co-interior = a, alternate = a + 40: $a + (a + 40) = 180$, $2a = 140$, $a = 70$, alternate = 110°. The 70° is the co-interior angle. The 90° needs a 0° difference. The 140° is 2a.

32. (C) Adjacent corner angles form a linear pair: $(3x + 18) + (2x + 12) = 180$, so $5x + 30 = 180$, $5x = 150$, $x = 30$. North-east = $3(30) + 18 = 108°$. The 72° is the north-west angle. The 30° is x. The 84° is a slip.

33. (A) Equal corresponding angles guarantee the two lines are parallel (the converse of the corresponding-angles rule). 'Perpendicular' needs a 90° angle, absent here. 'Nothing' ignores the valid converse. 'Meet at 102°' contradicts parallelism, since equal corresponding angles mean the lines do not meet.

34. (A) Co-interior angles sum to 180°, so angle 2 = 80°. The bisectors make 50° and 40° with the transversal on the same side; in the triangle they form with the transversal, the angle between the bisectors = $180 - 50 - 40 = 90°$. The bisectors of co-interior angles of parallel lines always meet at right angles. The 50° and 40° are the half-angles. The 100° just repeats angle 1.

35. (C) Sharing a vertex as a linear pair, they sum to 180°. Parts $5 + 4 = 9$, each 20°. Alternate-interior = $5 \times 20 = 100°$. The 80° is the co-interior partner (4 parts). The 90° needs a 1:1 ratio. The 108° comes from a 3:2 split, the wrong ratio.

36. (B) For parallel lines co-interior angles must sum to exactly 180°. Here 95 + 86 = 181° ≠ 180°, so the lines are NOT parallel, missing by 1°. 'Parallel' wrongly claims a 180° sum. 'Perpendicular' misreads a near-90° angle as decisive. 'Differ by 9°' uses the difference (the wrong test) instead of the sum.

37. (C) Alternate interior angles are equal: 2x + 25 = 3x + 5, so x = 20. Angle QAB = 2(20) + 25 = 65°. The 20° is x. The 115° is the supplement (a co-interior trap). The 85° is 3x + 25, a slip.

38. (A) Corresponding angles are equal: x + 40 = 3x − 20, so 60 = 2x, x = 30. The angle = 30 + 40 = 70°. Its alternate-interior angle (parallel lines) equals it: 70°. The 110° is the supplement (a co-interior trap). The 30° is x. The 50° is a slip.

39. (B) The 142° exterior angle has a linear-pair interior neighbour of 180 − 142 = 38° at the same crossing. By alternate interior angles (parallel lines), the matching interior angle on the lower line equals that 38°. The 142° forgets the linear-pair step. The 52° is 90 − 38, unjustified. The 48° is an arithmetic slip.

40. (A) Reading around, the angles alternate a, b, a, b. The two x angles are the vertically opposite pair, and (2x − 30) is adjacent to x, forming a linear pair: x + (2x − 30) = 180, 3x = 210, x = 70. The 50 and 60 fail the linear-pair equation. The 75 comes from 3x = 225, a slip.

41. (A) Angle BOD = angle AOC = 52° (vertically opposite). Angle AOE = 90° means OE is perpendicular to AB, so angle EOB = 90°. With D and E on the same side of AB, angle EOD = angle EOB − angle DOB = 90 − 52 = 38°. The 52° forgets the 90° subtraction. The 128° is 90 + 38, wrong. The 90° ignores D.

42. (A) Sharing a vertex as a linear pair, they sum to 180°. Let co-interior = a, alternate = 2a: a + 2a = 180, 3a = 180, a = 60, alternate = 120°. The 60° is the co-interior angle. The 90° needs a 1:1 ratio. The 135° comes from a 3:1 ratio, not 2:1.

43. (C) Two intersecting lines give vertically opposite equal angles and linear pairs summing to 180°, so going around the pattern must be a, b, a, b with a + b = 180. 70,110,70,110 fits. The 70,110,110,70 ordering puts equal angles adjacent, breaking the alternating pattern. 90,90,90,90 is perpendicular (excluded). 70,70,110,110 also misorders the pattern.

44. (C) The two values form linear pairs, so they are supplementary: a + 2a = 180, 3a = 180, a = 60°. The smaller is 60°, the larger 120°. The 120° is the larger. The 90° needs equal values. The 45° comes from 180/4, a wrong split.

45. (D) Co-interior angles are supplementary. Let smaller = a, larger = a + 24: a + (a + 24) = 180, 2a = 156, a = 78, larger = 102°. The 78° is the smaller. The 90° needs a 0° difference. The 114° comes from adding 24 to 90 by mistake.

46. (B) Alternate angles between parallel lines are equal, so angle A = angle B. Substituting: angle A = 2·angle A − 24, so 24 = angle A, i.e. angle A = 24°. The 48° doubles it. The 12° halves it. The 156° is the supplement, irrelevant for equal alternate angles.

47. (A) A perpendicular transversal makes every one of the eight angles 90°, so there is exactly ONE distinct value, 90°. '2 values: 90 and 90' lists the same value twice. '4 distinct values' arises only for an oblique unequal arrangement, not here. '80 and 100' is a generic oblique answer, not the perpendicular case.

48. (A) Reading around, the angles alternate a, b, a, b. The two (3x) angles are the vertically opposite pair, and (x + 20) is adjacent to 3x, forming a linear pair: 3x + (x + 20) = 180, 4x = 160, x = 40. The 20 and 35 fail the equation. The 45 comes from 4x = 180, a slip.

49. (B) Sharing a vertex as a linear pair, they sum to 180°. Parts 7 + 2 = 9, each 20°. The smaller (co-interior, 2 parts) = 2×20 = 40°. The 140° is the larger alternate-interior angle (7 parts). The 90° needs a 1:1 ratio. The 20° is one part, not two.

50. (D) Angle 2 = angle 1 = 124° (corresponding). Angle 3 = 180 − 124 = 56° (linear pair with angle 2). Angle 4 = angle 3 = 56° (vertically opposite). The 124° stops at angle 2. The 90° is unjustified. The 34° is 124 − 90, a wrong step.

51. (A) Parallel lines force corresponding angles to be equal. Here 88° ≠ 92°, so the lines are NOT parallel. 'Parallel' applies the rule without checking the angles match. 'Perpendicular' needs a 90° angle exactly, absent. 'Coincident' would make all angles equal, contradicting the data.

52. (A) Corresponding angles are equal: x + 25 = 2x − 15, so x = 40. The angle = 40 + 25 = 65°. Its co-interior partner is supplementary: 180 − 65 = 115°. The 65° is the corresponding angle itself, not its co-interior partner. The 40° is x. The 130° doubles 65 instead of subtracting from 180.

53. (A) Sharing a vertex as a linear pair, they sum to 180°. Parts 3 + 2 = 5, each 36°. Alternate-interior (3 parts) = 3×36 = 108°. The 72° is the co-interior partner (2 parts). The 90° needs a 1:1 ratio. The 120° comes from a 2:1 ratio.

54. (B) Vertically opposite: 4x + 5 = 6x − 25, so 30 = 2x, x = 15. Angle AOC = 4(15) + 5 = 65°. Its linear-pair neighbour AOD = 180 − 65 = 115°. The 65° is angle AOC itself, not its neighbour. The 15° is x. The 130° doubles 65 instead of subtracting from 180.

55. (C) The 84° corresponding angle at line a equals the corresponding angle at line b (84°). Its bisector splits it into 42° and 42°, so the bisector makes 42° with line b. The 84° forgets the bisection. The 48° is 90 − 42, unjustified. The 138° is the obtuse supplement, not the acute angle.

56. (D) Alternate interior angles are equal: $2x + 9 = 3x - 6$, so $x = 15$. Each angle = $2(15) + 9 = 39°$. The 15° is x. The 141° is the supplement, wrong because alternate angles are equal. The 45° is 3x, a slip.

57. (D) Angle Q is co-interior to angle P, so angle Q = $180 - 73 = 107°$. Its bisector splits it into 53.5° each, so the bisector makes 53.5° with line m. The 36.5° halves angle P instead. The 73° forgets angle Q and the bisection. The 107° is angle Q itself, not the bisected half.

58. (C) Linear pair: $7x + 2x = 180$, $9x = 180$, $x = 20$. Angle AOC = $7(20) = 140°$. OD bisects it, so angle DOC = $140/2 = 70°$. The 140° is the whole angle AOC, not its half. The 40° is angle COB (2x), not the bisected half. The 20° is x.

59. (C) Co-interior angles are supplementary. Let the smaller be a and the larger 4a: $a + 4a = 180$, $5a = 180$, $a = 36$, larger = $4(36) = 144°$. The 36° is the smaller. The 90° wrongly assumes equality. The 135° would come from a 3:1 ratio, not 4:1.

60. (D) The 118° angle at P has a corresponding angle at Q equal to 118° (parallel lines). The angle vertically opposite that corresponding angle is also 118° (vertically opposite angles are equal). So the required angle is 118°. The 62° is the supplement, the trap of inserting a linear-pair step that is not needed. The 90° is unjustified. The 236° is 2×118, a nonsense doubling.